

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397241
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Huang; Jiayang

Children's car with cleaning function

Abstract

The invention provides a children's car with cleaning function, comprising: a car body provided with a storage portion; a connecting frame, whose one end is connected with the car body, and the other end is provided with a connecting portion; and a cleaning tool head detachably connected with the connecting portion. The connecting frame includes an expanded state and a folded state; when the connecting frame is in the expanded state, the connecting frame expands to an outer side of the car body; when the connecting frame is in the folded state, the connecting portion is separated from the cleaning tool head and the connecting frame is located on the storage portion. When the cleaning function is not used, the cleaning tool head is detached from the connecting frame, and the connecting frame is folded to the storage portion, facilitating children to drive children's cars for fun.

Inventors:	Huang; Jiayang (Fujian, CN)
Applicant:	Huang; Jiayang (Fujian, CN)
Family ID:	1000008545032
Appl. No.:	19/090362
Filed:	March 25, 2025

Publication Classification

Int. Cl.: A63H17/38 (20060101); A47L9/00 (20060101); A47L13/24 (20060101)

U.S. Cl.:

CPC A63H17/38 (20130101); A47L9/009 (20130101); A47L13/24 (20130101);

Field of Classification Search

CPC: A47L (25/00); A47L (9/009); A47L (13/24); A47L (13/12); B62K (9/00); E01H (1/045)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1933501	12/1932	Allen	446/411	A63H 33/305
2703897	12/1954	Malm	15/48	A47L 11/4072
3197918	12/1964	Coggeshall	446/290	A63H 7/04
3354489	12/1966	Ehrlich	15/83	B66F 9/18
3959922	12/1975	Leistikow	446/144	A63H 33/30
4031659	12/1976	Keller	446/427	A63H 17/05
4595380	12/1985	Magers	446/451	A63H 17/38
4917648	12/1989	Hartje	446/427	A63H 17/05
5378191	12/1994	Ryaa	446/144	A63H 33/305
5435768	12/1994	Dunleavy	446/427	A63H 17/05
5920939	12/1998	Worwag	15/83	A47L 9/0411
6264528	12/2000	Doan	446/466	A63H 17/06
6272946	12/2000	Roux	74/498	A63H 17/38
6405817	12/2001	Huntsberger	280/47.11	B62K 9/00
6485351	12/2001	Bruder	446/427	A63H 17/05
6619677	12/2002	Wei	280/87.021	B62K 9/00
6949000	12/2004	Antoine	446/237	A63H 33/007
6968587	12/2004	Grey	15/377	A47L 5/30
7143461	12/2005	Spooner	15/41.1	A47L 11/4072
7175501	12/2006	Choi	446/469	B62K 9/00
7527293	12/2008	Wang	280/771	B62D 1/10
7794303	12/2009	Cassidy	446/460	A63H 33/30
7967317	12/2010	Ewringmann	280/288.4	E01H 1/056
8469369	12/2012	McCarthy	280/282	B62K 9/00
10265637	12/2018	Bernbaum	N/A	B62B 7/12
11918932	12/2023	Marlor	N/A	A63H 33/30
2004/0034952	12/2003	Ho	15/48.1	A47L 11/4041
2009/0203293	12/2008	Ewringmann	15/88.4	E01H 1/056
2013/0029560	12/2012	Mayson	446/468	A63H 17/38
2014/0099861	12/2013	Mayson	446/468	A63H 17/36
2023/0414051	12/2022	Foss	N/A	A47L 9/242
2024/0000276	12/2023	Millington	N/A	A47L 9/165
2024/0292990	12/2023	Hamilton	N/A	A47L 9/2857
2024/0375702	12/2023	Cui	N/A	B62D 1/10
2025/0083062	12/2024	Cui	N/A	A63H 17/38

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202847908	12/2012	CN	A47L 13/12

OTHER PUBLICATIONS

Primary Examiner: Kim; Eugene L

Assistant Examiner: Stanczak; Matthew B

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to the technical field of children's cars, and particularly relates to a children's car with cleaning function.

BACKGROUND

(2) A children's car is a kind of toy car driven by an electric motor that children can drive. It not only helps to exercise children's physical abilities, but also enhances their intelligence and allows them to experience the fun of being "drivers". At present, most of electric toy cars for children on the market only provide simple recreational functions, making their functions relatively limited. Some children's toy cars are provided with cleaning tools on car bodies, and the car bodies are connected to the cleaning tools through connecting frames, so that the children can experience the labor of cleaning the ground while having fun. However, the cleaning tools on these children's toy cars cannot be disassembled, making it inconvenient to store the cleaning tools and the connecting frames when the cleaning function is not used.

SUMMARY

(3) The present invention provides a children's car with cleaning function, aiming to solve the problem that the cleaning tools on the children's cars in the existing technology cannot be disassembled, making it inconvenient to store the cleaning tools and the connecting frames used for connecting the cleaning tools to the car bodies when the cleaning function is not used.

(4) In order to achieve the above technical problems, the present invention adopts the following technical solutions.

(5) A children's car with cleaning function, comprises a car body, which is provided with a storage portion; a connecting frame, whose one end of the connecting frame is connected with the car body, and the other end is provided with a connecting portion; and a cleaning tool head, which can be disassembly connected with the connecting portion; wherein the connecting frame includes an expanded state and a folded state; when the connecting frame is in the expanded state, the connecting frame expands to an outer side of the car body; when the connecting frame is in the folded state, the connecting portion is separated from the cleaning tool head and the connecting frame is located on the storage portion.

(6) As a further solution, when the connecting frame is in the expanded state, the connecting frame expands to a front side of the car body, and the cleaning tool head is located on the front side of the car body.

(7) Based on the above technical solution, the above settings make it convenient for the children to observe while driving the children's cars forward.

(8) As a further solution, there are two connecting frames, namely a left connecting frame and a right connecting frame; the left connecting frame is located on a left side of the car body, and the right connecting frame is located on a right side of the car body.

(9) There are two storage portions, namely a left storage portion and a right storage portion; the left storage portion is located on the left side of the car body, and the right storage portion is located on

the right side of the car body.

(10) When the connecting frame is in the folded state, the left connecting frame is located on the left storage portion, and the right connecting frame is located on the right storage portion.

(11) Based on the above technical solution, the two connecting frames are arranged and folded correspondingly on the two storage portions, so as to facilitate users' operations.

(12) As a further solution, wherein the storage portion is configured as a storage slot, and when the connecting frame is in the folded state, the connecting frame is rotated into the storage slot.

(13) As a further solution, the cleaning tool head is provided with a connecting base, and the connecting base is provided with two slots, namely a left slot and a right slot. The connecting portion on the left connecting frame is detachably inserted into the left slot through a bolt, and the connecting portion on the right connecting frame is detachably inserted into the right slot through the bolt.

(14) Based on the above technical solution, the above settings are convenient for the users to quickly install the cleaning tool head to the connecting frame or remove it from the connecting frame, and the two connecting portions are inserted into the two slots through the bolts respectively, so that the cleaning tool head is more stably connected with the car body.

(15) As a further solution, the connecting base is provided with a wrenching component, a middle part of the wrenching component is hinged on the connecting base through an articulated shaft, a wrenching portion is provided on a rear part of the wrenching component, and the left and right bolts are located on left and right sides of a front part of the wrenching component, respectively.

(16) When the wrenching portion is wrenched downward, the left and right bolts are separated from the left and right slots upwards, respectively; when the wrenching component is released, the left and right bolts are inserted downwards into the left and right slots, respectively.

(17) Based on the above technical solution, by setting the wrenching component, it is convenient for the users to operate and disassemble the cleaning tool head.

(18) As a further solution, the connecting frame is rotatably connected with the car body through a rotating shaft.

(19) Based on the above technical solution, the connecting frame is rotatably connected with the car body, which is convenient to switch the connecting frame between the expanded state and the folded state.

(20) As a further solution, the car body is provided with an accommodating groove, and an opening of the accommodating groove is detachably provided with a cover.

(21) Based on the above technical solution, by setting the accommodating groove, the cleaning tool head can be placed into the accommodating groove when the cleaning function is not used, which is convenient for storage.

(22) As a further solution, the cleaning tool head is a mop head or a vacuum cleaner head.

(23) As a further solution, a handle is detachably provided on the cleaning tool head.

(24) Based on the above technical solution, when the cleaning tool head needs to be detached from the car body for individual use, the handle can be assembled onto the cleaning tool head to facilitate operation; when the cleaning tool head needs to be assembled to the car body for use, the handle can be detached from the cleaning tool head to facilitate the assembly operation.

(25) As a further solution, wherein left and right ends of the handle are inserted into the slots and secured with the bolts.

(26) As a further solution, a steering wheel is detachably provided on the car body.

(27) Based on the above technical solution, the steering wheel can be disassembled, making it convenient to package, transport, and store the children's car.

(28) The present invention has the following the beneficial effects.

(29) The present invention provides a storage portion, a connecting frame and a cleaning tool head on a car body, and the cleaning tool head is detachably connected with the connecting portion. The connecting frame includes an expanded state and a folded state, so that when the cleaning function

needs to be used, the connecting frame can be switched to the expanded state, thus to facilitate to connect the cleaning tool head, therefore children can experience the labor of cleaning the ground while driving the children's cars for fun. When the cleaning function doesn't need to be used, the cleaning tool head can be detached from the connecting frame, and the connecting frame can be folded on the storage portion, so as to facilitate children to drive the children's cars for fun. In addition, the cleaning tool head can be detached from the connecting frame, which is convenient for packaging and transportation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to illustrate the technical solutions in the embodiments of the present invention more clearly, the drawings required in the description of the embodiments will be briefly introduced below. It should be understood that the drawings described in the following are merely some embodiments of the present invention, and therefore should not be regarded as limitations of the scope. For those skilled in the art, other drawings can also be obtained according to the drawing without making any creative efforts.

(2) FIG. 1 is a structural schematic view of the children's car with cleaning function according to the present invention, showing a connecting frame connected to a mop head in a first orientation.

(3) FIG. 2 is a structural schematic view of the children's car with cleaning function according to the present invention, showing the connecting frame connected to the mop head in a second orientation.

(4) FIG. 3 is a structural schematic view of the children's car with cleaning function according to the present invention, showing the connecting frame connected to a vacuum cleaner head.

(5) FIG. 4 is a structural schematic view of the connecting frame in a folded state.

(6) FIG. 5 is a structural schematic view that the connecting frame is connected to a cleaning tool head.

(7) FIG. 6 is an exploded structural view of a car body and the cleaning tool head.

(8) FIG. 7 is a structural schematic view that the vacuum cleaner head is connected to a handle.

(9) The reference numbers in the drawings are as follows: car body **1**; storage portion **10**; left storage portion **10a**; right storage portion **10b**; steering wheel **11**; cover **12**; front wheel **13**; rear wheel **14**; storage slot **15**; rotating shaft **16**; accommodating groove **17**; mounting seat **18**; connecting frame **2**; left connecting frame **2a**; right connecting frame **2b**; connecting portion **21**; mop head **3**; vacuum cleaner head **4**; connecting base **5**; cleaning tool head **50**; articulated shaft **51**; wrenching component **52**; wrenching portion **52a**; bolt **53**; left bolt **53a**; right bolt **53b**; slot **54**; left slot **54a**; right slot **54b**; handle **6**.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) The technical solutions in the embodiments of the present invention will be illustrated in detail in combination with the embodiments and the drawings. It should be understood that the specific embodiments described here are only used to explain the present invention and are not used to limit the present invention. Based on the embodiments of the present invention, all other embodiments obtained by a person skilled in the art without making creative efforts shall fall within the scope of the present invention.

(11) As shown in FIG. 1 to FIG. 5, the present embodiment provides a children's car with cleaning function, which comprises: a car body **1**, which is provided with a storage portion **10**; a connecting frame **2**, whose one end is connected with the car body **1**, and the other end is provided with a connecting portion **21**; and a cleaning tool head **50**, which is detachably connected with the connecting portion **21**.

(12) The connecting frame **2** includes an expanded state and a folded state, when the connecting

frame **2** is in the expanded state, the connecting frame **2** expands to an outer side of the car body **1**, that is, to a front, rear, left or right side of the car body **1**; when the connection frame **2** is in the folded state, the connecting portion **21** is separated from the cleaning tool head **50** and the connecting frame **2** is located on the storage portion **10**. Specifically, when the cleaning function of the children's car is not used, the cleaning tool head **50** is detached first, and then the connecting frame **2** is folded in the storage portion **10**.

(13) As shown in FIG. **1** to FIG. **4**, there are two front wheels **13** located on the left and right underneath a front side of the car body **1**, and there are two rear wheels **14** located on left and right underneath a rear side of the car body **1**. The car body **1** is provided with a steering wheel **11** and a riding portion **12** for children to ride and operate the car body **1**. The steering wheel **11** can be configured to be detachably connected to the car body **1**, so that the steering wheel **11** can be detached during packaging and transportation to facilitate easier packaging. Specifically, the steering wheel **11** is detachably connected to the car body **1** by means of a plug-in structure.

(14) Specifically, the cleaning tool head **50** can be a mop head **3**, a vacuum cleaner head **4**, etc. In specific use, users can choose to assemble either the mop head **3** or the vacuum cleaner head **4** onto the connecting frame **2**.

(15) In some specific embodiments, as shown in FIG. **1** to FIG. **3**, to facilitate the children's observation of the cleaning tool head while driving the children's car forward, a preferred solution of the present embodiment is that when the connecting frame **2** is in the expanded state, the connecting frame **2** is rotatably or slidably expanded to the front side of the car body **1**, and the cleaning tool head **50** is located on the front side of the car body **1**, and the cleaning tool head **50** is connected with the connecting portion **21** on the front side of the car body **1**.

(16) In some specific embodiments, as shown in FIG. **1** to FIG. **4**, the connecting frame **2** is rotatably connected to the car body **1** through a rotating shaft **16**. Specifically, a mounting seat **18** is provided on the car body **1**, and the connecting frame **2** is rotatably connected to the mounting seat **18** through the rotating shaft **16**. When the connecting frame **2** needs to be switched between the expanded state and the folded state, a quick switch can be achieved by rotating the connecting frame **2**.

(17) In some specific embodiments, as shown in FIG. **1** to FIG. **3**, there are two connecting frames **2**, namely a left connecting frame **2a** and a right connecting frame **2b**. The left connecting frame **2a** is located on a left side of the car body **1**, and the right connecting frame **2b** is located on a right side of the car body **1**.

(18) There are two storage portions, namely a left storage portion **10a** and a right storage portion **10b**. The left storage portion **10a** is located on the left side of the car body **1**, and the right storage portion **10b** is located on the right side of the car body **1**.

(19) When the connecting frame **2** is in the folded state, the left connecting frame **2a** is located on the left storage portion **10a**, and the right connecting frame **2b** is located on the right storage portion **10b**. Specifically, to ensure a more stable connection between the cleaning tool head **50** and the car body **1**, the connecting portions **21** of the two connecting frames **2** are detachably connected to left and right sides of the cleaning tool head **50**, respectively. Alternatively, it can also be set that the connecting portion **21** of each connecting frame **2** is separately connected to one of the cleaning tool head **50**.

(20) In some specific embodiments, as shown in FIG. **4**, in order to prevent the connecting frame **2** from occupying additional space when in the folded state, the storage portion **10** is configured as a storage slot **15** in this embodiment, and when the connecting frame **2** is in the folded state, the connecting frame **2** is rotated into the storage slot **15**.

(21) In some specific embodiments, as shown in FIG. **5**, to facilitate the user to assemble the cleaning tool head **50** to or detach it from the connecting frame **2**, the cleaning tool head **50** is provided with a connecting base **5**, and the connecting base **5** is provided with two slots **54**, namely a left slot **54a** and a right slot **54b**. The connecting portion **21** on the left connecting frame **2a** is

detachably inserted into the left slot **54a** via a left bolt **53a**, and the connecting portion **21** on the right connecting frame **2b** is detachably inserted into the right slot **54b** via a right bolt **53b**. Specifically, each connecting portion **21** is provided with a bolt hole, and the bolt **53** is inserted from top to bottom through a top wall of the slot **54** and into the bolt hole, thereby firmly fixing the cleaning tool head **50** to the connecting base **5**.

(22) As shown in FIG. 5, in order to facilitate user's operations, the connecting base **5** is provided with a wrenching component **52**, a middle part of the wrenching component **52** is hinged on the connecting base **5** through an articulated shaft **51**, a wrenching portion is provided on a rear part of the wrenching component **52**, and the left and right bolts **53a/53b** are respectively located on left and right sides of a front part of the wrenching component **52**.

(23) When the wrenching portion is wrenched downward, the left and right bolts **53a/53b** are respectively separated from the left and right slots **54a/54b** upwards and the two connecting portions **21**, so that the cleaning tool head **50** can be detached from the two connecting frames **2**. When the wrenching portion is released, the left and right bolts **53 a/53b** are inserted downward into the left and right slots **54 a/54b**, respectively.

(24) In some specific embodiments, as shown in FIG. 7, for the convenience of users, a handle **6** can be detachably provided on the cleaning tool head **50**. A specific method is that a connection between the cleaning tool head **50** and the handle **6** is the same as that between the cleaning tool head **50** and the connecting frame **2**, that is, left and right ends of the handle **6** are inserted into the slots **54** and secured with the bolts **53**. When the vacuum cleaner head **4** is connected to the car body **1** for use, the handle **6** on the vacuum cleaner head **4** is detached. When the vacuum cleaner head **4** is detached from the car body **1** for use, the handle **6** is assembled on the vacuum cleaner head **4**.

(25) In some specific embodiments, as shown in FIG. 6, after the cleaning tool head **50** is detached from the connecting frame **2**, in this embodiment, an accommodating groove **17** is provided on the car body **1** to facilitate the storage of the cleaning tool head **50**. An opening of accommodating groove **17** is detachably provided with a cover **12**. Specifically, the cover **12** is used for children to sit on. The accommodating groove **17** is provided below the cover **12** so as to make full use of the space.

(26) The working principle is described as follow.

(27) When the cleaning function of the children's car is used, the user can pull the connecting frame **2** out from the storage slot **15** and rotate it to the front side of the car body **1**, and then assemble the mop head **3** or the vacuum cleaner head **4** onto the connecting frame **2**. While the child drives the children's car forward, the ground is cleaned by the mop head **3** or the vacuum cleaner head **4**. When the cleaning function is not in used, the mop head **3** or the vacuum cleaner head **4** is detached from the connecting frame **2**, and then the connecting frame **2** is rotated back into the storage slot **15**.

(28) The present invention is not limited to the above optional implementation methods. Provided that there is no mutual conflict, various solutions can be combined arbitrarily. Anyone, under the inspiration of the present invention, may derive other forms of products. However, regardless of any changes made in their shapes or structures, any technical solutions that fall within the scope defined by the claims of the present invention shall be within the protection scope of the present invention.

Claims

1. A children's car with cleaning function, comprising a car body, which is provided with a storage portion; a connecting frame, whose one end is connected with the car body, and the other end is provided with a connecting portion; and a cleaning tool head, which is detachably connected with the connecting portion; wherein the connecting frame includes an expanded state and a folded state;

when the connecting frame is in the expanded state, the connecting frame expands to an outer side of the car body; when the connecting frame is in the folded state, the connecting portion is separated from the cleaning tool head and the connecting frame is located on the storage portion, wherein there are two connecting frames, namely a left connecting frame and a right connecting frame: the left connecting frame is located on a left side of the car body, and the right connecting frame is located on a right side of the car body; there are two storage portions, namely a left storage portion and a right storage portion: the left storage portion is located on the left side of the car body, and the right storage portion is located on the right side of the car body; when the connecting frame is in the folded state, the left connecting frame is located on the left storage portion, and the right connecting frame is located on the right storage portion, wherein the cleaning tool head is provided with a connecting base, and the connecting base is provided with two slots, namely a left slot and a right slot: the connecting portion on the left connecting frame is detachably inserted into and connected with the left slot via a left bolt, and the connecting portion on the right connecting frame is detachably inserted into and connected with the right slot via a right bolt.

2. The children's car with cleaning function according to claim 1, wherein when the connecting frame is in the expanded state, the connecting frame expands to a front side of the car body, and the cleaning tool head is located on the front side of the car body.

3. The children's car with cleaning function according to claim 1, wherein the storage portion is configured as a storage slot, and when the connecting frame is in the folded state, the connecting frame is rotated into the storage slot.

4. The children's car with cleaning function according to claim 1, wherein the connecting base is provided with a wrenching component, a middle part of the wrenching component is hinged on the connecting base through an articulated shaft, a wrenching portion is provided on a rear part of the wrenching component, and the left and right bolts are located on left and right sides of a front part of the wrenching component, respectively; when the wrenching portion is wrenched downward, the left and right bolts are separated from the left and right slots upwards, respectively; when the wrenching component is released, the left and right bolts are inserted downwards into the left and right slots, respectively.

5. The children's car with cleaning function according to claim 1, wherein the connecting frame is rotatably connected with the car body through a rotating shaft.

6. The children's car with cleaning function according to claim 1, wherein the car body is provided with an accommodating groove, and an opening of the accommodating groove is detachably provided with a cover.

7. The children's car with cleaning function according to claim 1, wherein the cleaning tool head is a mop head or a vacuum cleaner head.

8. The children's car with cleaning function according to claim 1, wherein a handle is detachably provided on the cleaning tool head.

9. The children's car with cleaning function according to claim 1, wherein a steering wheel is detachably provided on the car body.
